AI PDF Chatbot using RAG and LLM

Name: Jayanth M

Course: M.Tech – Artificial Intelligence & Data Science

Institution: Alliance University

Abstract

This project presents an intelligent AI-based chatbot capable of answering user queries from PDF documents using Retrieval-Augmented Generation (RAG). The system extracts textual data from PDF files and processes it into smaller chunks, which are then converted into vector embeddings using pre-trained models. These embedding's are stored in a FAISS vector database for efficient similarity search. When a user asks a question, the system retrieves the most relevant content from the database and generates a meaningful response using a Large Language Model (GPT-2). This approach ensures that responses are grounded in the document content, improving accuracy and relevance. The project demonstrates practical implementation of modern AI techniques in document understanding and conversational systems.

Objectives

- To develop a chatbot capable of understanding PDF documents
- To implement Retrieval-Augmented Generation (RAG)
- To improve answer accuracy using document-based retrieval
- To integrate a Large Language Model for natural responses
- To build a real-world AI application for document Q&A

Introduction

With the rapid growth of digital documents, extracting meaningful information efficiently has become a significant challenge. Large volumes of data are stored in formats such as PDFs, reports, and research papers, making it difficult for users to quickly find relevant information. Traditional search methods are limited as they rely primarily on keyword matching and fail to understand the context or intent behind a user's query. This often results in incomplete or irrelevant search results.

Artificial Intelligence (AI), particularly Natural Language Processing (NLP), provides powerful techniques to overcome these limitations. NLP enables machines to understand, interpret, and generate human language in a meaningful way. Recent advancements in AI have led to the development of Large Language Models (LLMs), which are capable of generating human-like responses based on given inputs.

In this project, a modern approach called Retrieval-Augmented Generation (RAG) is used. RAG combines two important components: information retrieval and language generation. The retrieval component searches for relevant information from a document database, while the generation component uses a language model to produce accurate and context-aware responses. This combination improves both the accuracy and reliability of the system.

The proposed system is an AI-based chatbot that allows users to interact with PDF documents in a conversational manner. Instead of manually reading through entire documents, users can simply ask questions and receive precise answers instantly. The system processes the PDF, extracts and stores its content efficiently, and retrieves relevant information based on user queries before generating a response.

This approach has wide applications across various domains, including education, where students can quickly understand study materials; research, where large documents can be analyzed efficiently; and business environments, where quick insights from reports and documents are essential. Overall, this project demonstrates how AI can transform traditional document search into an intelligent, interactive, and user-friendly experience.

System Architecture / Methodology

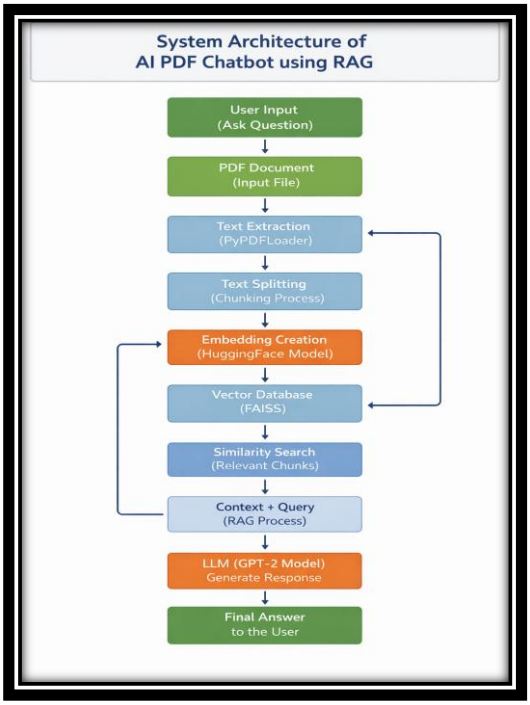


Fig : System Architecture

The system architecture of the AI PDF Chatbot using Retrieval-Augmented Generation (RAG) illustrates the complete workflow of how user queries are processed and answered using information extracted from PDF documents.

The process begins with the user providing a query, such as a question related to the content of a PDF document. Along with this, a PDF file is provided as input to the system, which serves as the knowledge source for generating responses.

The first step involves extracting text from the PDF document using a PDF loader. Since the extracted text may be large and unstructured, it is divided into smaller segments through a process called text splitting or chunking. This improves efficiency and ensures better retrieval of relevant information. Next, each text chunk is converted into numerical representations known as embeddings using a pre-trained model from HuggingFace. These embeddings capture the semantic meaning of the text and allow the system to understand the content in a meaningful way.

The generated embeddings are then stored in a vector database called FAISS. This database enables fast and efficient similarity search, allowing the system to quickly find relevant information based on user queries. When the user inputs a query, the system converts the query into an embedding and performs a similarity search in the FAISS database. The most relevant text chunks are retrieved based on their similarity to the query.These retrieved text chunks are combined with the user query in a process known as Retrieval-Augmented Generation (RAG). This ensures that the response is grounded in the actual content of the document rather than being generated randomly.

Finally, the combined input is passed to a Large Language Model (GPT-2), which generates a natural language response. The generated answer is then displayed to the user. This architecture effectively combines information retrieval and language generation, resulting in accurate and context-aware responses from PDF documents.

Tools and Technologies Used

- **Programming Language:** Python
- **Framework:** LangChain
- **Vector Database:** FAISS
- **Embedding Model:** Sentence Transformers
- **LLM:** GPT-2 (HuggingFace Transformers)
- **Platform:** Google Colab
- **Libraries:** PyPDF, Transformers, NumPy

Key Features

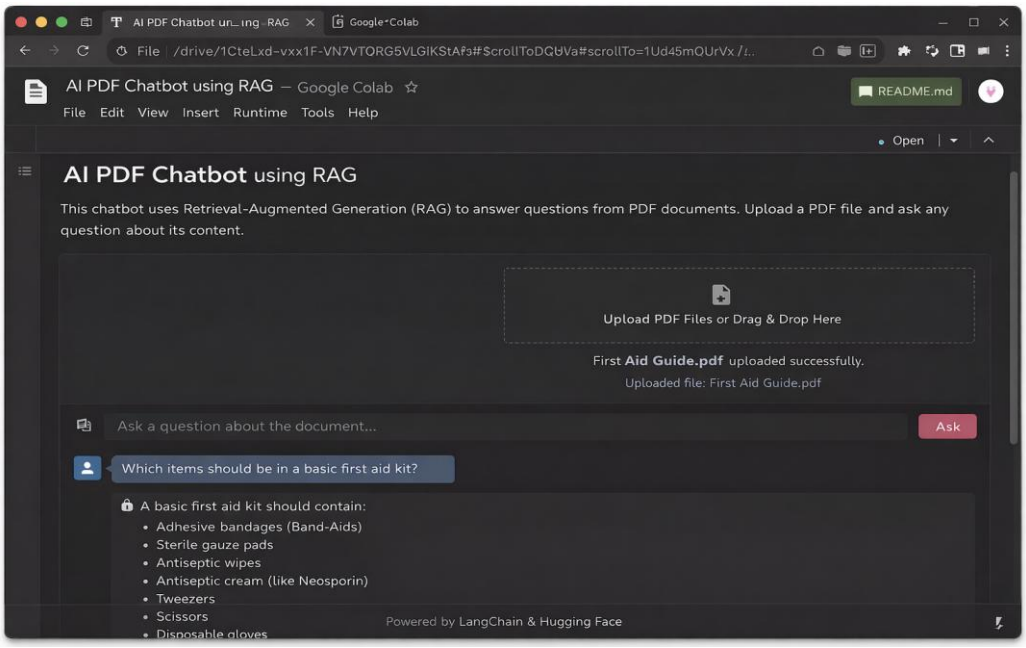
- Upload and process PDF documents
- Extract and analyze text automatically
- Intelligent question answering system
- Fast retrieval using vector database
- Context-aware responses using LLM
- Easy to extend and scalable

Results and Output

The system successfully:

- Extracts text from PDF files
- Retrieves relevant information based on user queries
- Generates meaningful responses

Although GPT-2 provides basic responses, the system demonstrates the effectiveness of combining retrieval and generation techniques.



Limitations

- GPT-2 generates less accurate responses compared to advanced models
- Performance depends on quality of PDF content

Future Enhancements

- Integrate advanced LLMs (ChatGPT, LLaMA, Gemini)
- Develop web interface using Streamlit
- Add support for multiple documents
- Improve accuracy using fine-tuned models
- Deploy as a web application

Conclusion

This project demonstrates how AI can be effectively used to build intelligent document-based chat systems. By combining Retrieval-Augmented Generation with Large Language Models, the system provides accurate and context-aware answers from PDF documents. It highlights the potential of modern AI techniques in solving real-world problems related to information retrieval and understanding.